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20 HOME : PRINT : PRINT
30 DEF FN ASN(X) = ATN (X / SQR ( - X * X + 1))
40 DEF FN ACO(X) = - ATN (X / SQR ( - X * X + 1)) + 1.5707963
50 DEF FN RAD(X) = .01745328 * (X)
60 DEF FN DEG(X) = 57.29578 * (X)
70 REM INSERT LOCAL LONGITUDE AND LATITUDE IN FOLLOWING STATEMENT
80 LO = 122.49:LA = 38.24
90 PRINT : PRINT : PRINT : PRINT
100 PRINT TAB(10)"ASTRONOMY PROGRAM": PRINT
110 PRINT TAB(9)"-----"
120 PRINT TAB(9)"1 INNER PLANETS 1"
130 PRINT TAB(9)"1 TIMES FROM SUN 1"
140 PRINT TAB(9)"-----"
150 PRINT
160 PRINT TAB(7)"BY ERIC BURGESS F.R.A.S."
170 PRINT
180 PRINT TAB(7)"ALL RIGHTS RESERVED BY"
190 PRINT TAB(7)"S & T SOFTWARE SERVICE"
200 FOR J = 2000 TO 1 STEP -1: NEXT J
210 HOME : PRINT : PRINT : PRINT
220 PRINT : PRINT : PRINT : PRINT
230 PRINT "THIS PROGRAM GIVES APPROXIMATE"
240 PRINT "TIME IN MINUTES THAT MERCURY AND"
250 PRINT "VENUS RISE BEFORE THE SUN AS"
260 PRINT "MORNING STARS, OR SET AFTER THE"
270 PRINT "SUN AS EVENING STARS, FOR ANY DATE"
280 PRINT " WHICH YOU INPUT"
290 PRINT : PRINT
300 PRINT "INITIAL CONDITIONS ARE SET FOR....."
310 PRINT TAB(5)"SEBASTOPOL, CA LONG. 122.49"
320 PRINT TAB(21)"LAT. 38.24": PRINT
330 PRINT : INPUT "WANT TO CHANGE THEM Y/N? ";A$
340 IF A$ < > "Y" GOTO 380
350 HOME : PRINT : PRINT : PRINT : PRINT
360 INPUT " GIVE LATITUDE ";LA
370 INPUT " AND LONGITUDE ";LO
380 HOME : PRINT : PRINT : PRINT
390 PRINT "ENTER THE DATE": PRINT
400 FL = 2
410 INPUT " THE YEAR ? ";YD$:Y = VAL (YD$)
420 IF Y = 0 THEN PRINT "INVALID RESPONSE": PRINT : GOTO 410
430 IF Y > 1800 THEN GOTO 490
440 PRINT "IS ";Y;" THE CORRECT YEAR? "
450 INPUT ;YS
460 IF Y$ = "Y" THEN GOTO 490
470 IF Y$ < > "N" THEN PRINT "INVALID RESPONSE": PRINT : GOTO 440
480 IF Y$ = "N" THEN PRINT : GOTO 410
490 LY = 0
500 IF Y / 4 = INT (Y / 4) = 0 AND Y / 100 = INT (Y / 100) < > 0 THEN
LY = 1
510 PRINT : INPUT " THE MONTH? ";MD$:M = VAL (MD$)
520 IF M = 0 OR M > 12 THEN PRINT "INVALID RESPONSE": PRINT : GOTO 490

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